Isolation Forest Outliers: Numerical Walkthrough

Isolation Forest isolates anomalies directly by recursively slicing the data space with random lines. Because anomalies are sparse and far away from the crowd, they get isolated in very few slices (shallow tree depth), whereas normal points require many slices to separate.

The Sample Dataset

Imagine we are auditing Credit Card Transactions with two features: Spend Amount ($) and Distance from Home (km). We have 4 transactions (3 normal, 1 outlier):

- **Point A**: Spend = $20, Distance = 2 km (Normal)
- **Point B**: Spend = $25, Distance = 5 km (Normal)
- **Point C**: Spend = $30, Distance = 3 km (Normal)
- **Point X**: Spend = $500, Distance = 80 km (Anomaly Outlier)

Step-by-Step Space Slicing

Round 1: The First Split
1. Pick random feature: The algorithm randomly chooses **Spend Amount**.
2. Find range: The minimum spend is $20 (Point A) and maximum is $500 (Point X).
3. Pick random split value: It picks a random number between 20 and 500. Let's say it picks **$150**.
4. Divide the dataset:

- Left Branch (Spend < $150): Contains Point A, Point B, Point C.
- Right Branch (Spend ≥ $150): Contains Point X.

✨ **Point X is now completely alone. It is isolated in just 1 split! (Path Length = 1)**

Round 2: Splitting the Left Branch

We look only at the remaining group {A, B, C} on the left branch:

1. Pick random feature: It randomly chooses **Distance from Home**.
2. Find range: Min = 2 km, Max = 5 km.
3. Pick random split value: It picks a random number between 2 and 5. Let's say it picks **4 km**.
4. Divide the remaining dataset:

- Left Sub-Branch (Distance < 4 km): Contains Point A (2 km) and Point C (3 km).
- Right Sub-Branch (Distance ≥ 4 km): Contains Point B (5 km).

✨ **Point B is now completely alone. It is isolated after 2 splits! (Path Length = 2)**

Round 3: Splitting the Last Group

Only Point A and Point C are left together:

1. Pick random feature: It chooses **Spend Amount**.
2. Find range: Min = $20, Max = $30.
3. Pick random split value: It randomly picks **$26**.
4. Divide the dataset:

- Left Sub-Branch (Spend < $26): Contains Point A ($20).
- Right Sub-Branch (Spend ≥ $26): Contains Point C ($30).

✨ **Point A and Point C are now both isolated. (Path Length = 3 for both)**

Data Point	Transaction Details	Splits to Isolate (Path Length)	Verdict
Point X	Anomaly ($500, 80km)	**1 split** (Very Shallow)	🚨 **HIGH RISK FRAUD FLAG**
Point B	Normal ($25, 5km)	2 splits	✅ Approved (Normal)
Point A	Normal ($20, 2km)	3 splits (Deep)	✅ Approved (Normal)
Point C	Normal ($30, 3km)	3 splits (Deep)	✅ Approved (Normal)